

★ WEEK 1 · PART 3 · PRACTICE

# Market Efficiency, Worked by Hand

## Test the Random Walk With a Variance Ratio and a Runs Count

Weak-form efficiency, and the limits of testing it

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Part 3's central question, with the arithmetic shown. We run two classic weak-form tests - the variance ratio and the runs test - to ask whether past prices predict future returns, then confront why efficiency can never quite be proven: the joint hypothesis problem and the Grossman-Stiglitz paradox.

# 3P

## PART 3 · APPLICATIONS

### Can the Past Predict the Future?

Case A: the variance ratio test - under a random walk, variance scales linearly with the horizon, so a ratio above one signals trending. Case B: the runs test counts up-and-down streaks against what randomness predicts - and the deep reasons efficiency resists proof. Then a Claude Code lab.

# What Part 3 Told Us

Now we apply it - with numbers

1

## **The random walk makes a testable claim**

Tomorrow's price is today's plus drift plus an unpredictable shock. So return variance should scale linearly with the horizon.

2

## **Weak-form efficiency has classic tests**

Autocorrelation, the variance ratio, and the runs test all ask the same thing: can past prices predict future returns?

3

## **Efficiency can never be tested alone**

The joint hypothesis problem and the Grossman-Stiglitz paradox mean perfect efficiency is untestable - and impossible.

**Everything here is a calculator exercise: one variance ratio, one runs count.**

# How These Cases Work

Data, then arithmetic, then meaning

## Each case has three steps

- Start from returns or their signs.
- Compute the ratio, or count the runs.
- Then judge efficiency.

## What you should watch

- How variance should scale under a random walk.
- How the runs count compares to its expectation.
- Why a clean verdict is so hard to reach.

# The Variance Ratio Test

Does the market follow a random walk?

**Under a random walk, returns are independent, so the variance of a k-period return is exactly k times the one-period variance. The variance ratio divides the observed multi-period variance by that prediction. A ratio of one means a random walk; above one signals trending, below one mean reversion. We compute it for a five-day**

## The inputs

- One-day return variance of 4.
- Observed five-day variance of 26.
- The random-walk prediction is 20.

## What we'll do by hand

- Form the ratio of observed to predicted.
- Read whether it exceeds one.
- Interpret the market structure.

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# The Random Walk's Prediction

Variance scales linearly

$$P_t = P_{t-1} + \mu + \varepsilon_t \Rightarrow \text{Var}(r_k) = k \cdot \text{Var}(r_1) \quad (\text{variance scales linearly with the horizon})$$

## Why linear scaling

- Independent shocks add their variances.
- So k periods give k times the variance.
- That's the null the test checks.

## The ratio

- Observed variance over k times the one-period.
- Exactly one under a random walk.
- Deviations reveal autocorrelation.

# Computing the Ratio

Observed over predicted

$$VR(5) = \frac{\text{Var}(r_5)}{5 \cdot \text{Var}(r_1)} = \frac{26}{5 \times 4} = \frac{26}{20} = 1.30$$

## The calculation

- The random walk predicts five times four, or 20.
- The observed five-day variance is 26.
- The ratio is 26 over 20, or 1.30.

## First read

- The ratio exceeds one.
- Variance grew faster than linearly.
- That points to positive autocorrelation.

# Reading the Ratio

Trending, random, or reverting

$VR = 1$  random walk;  $VR > 1$  trending;  $VR < 1$  mean-reverting

## What 1.30 means

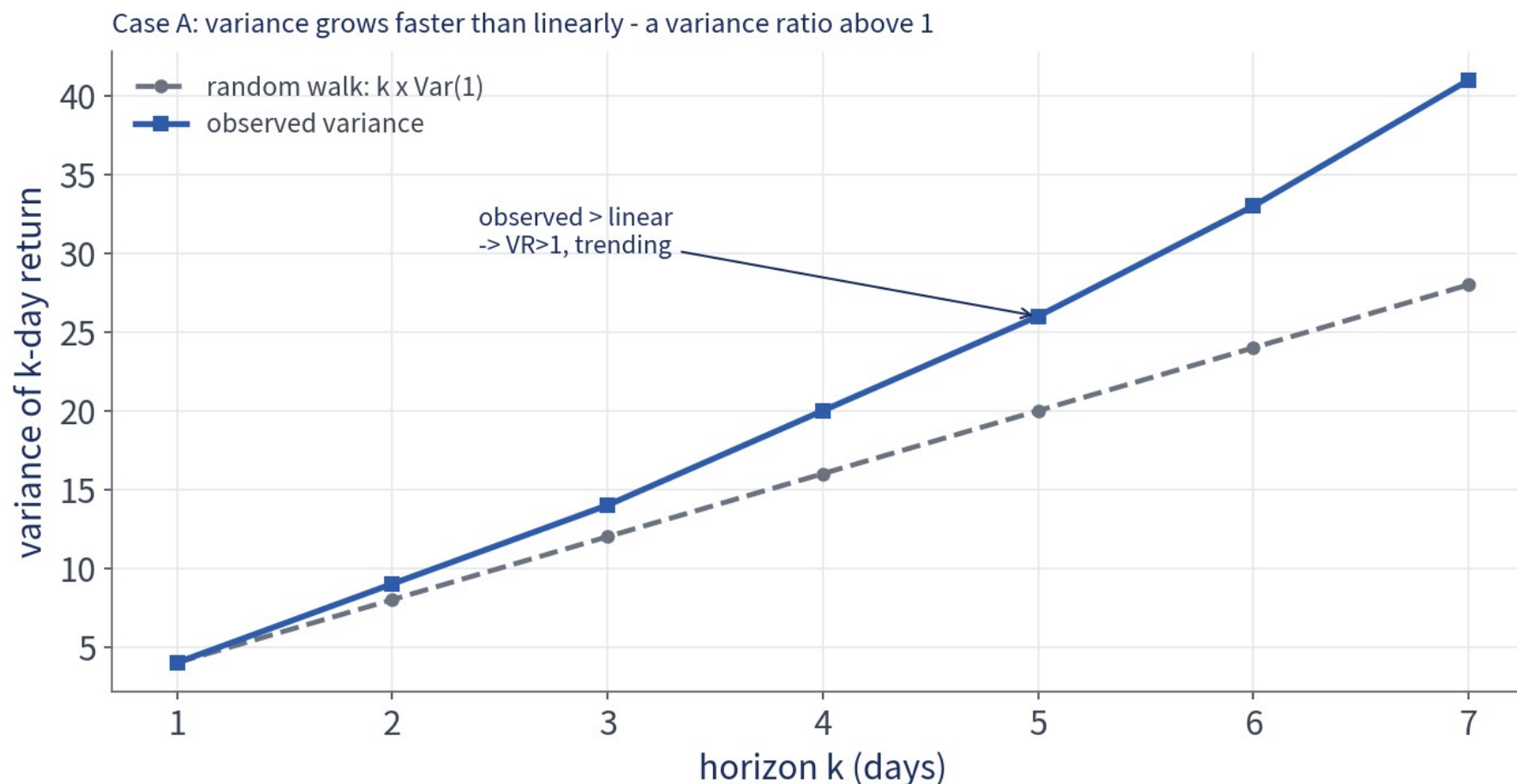
- Above one: returns trend.
- Gains tend to follow gains.
- A mild momentum, a mild inefficiency.

## The other cases

- Exactly one: a pure random walk.
- Below one: mean reversion.
- The ratio maps to net autocorrelation.

# Faster Than Linear

The dashed line is the random walk's prediction: variance rising in exact proportion to the horizon. The observed variance climbs above it, curving upward - the signature of positive autocorrelation. That gap is the variance ratio exceeding one, and it says the series trends rather than wandering purely at random.



# Three Forms of Efficiency

How much is already priced in

## The three forms

- Weak: past prices are priced in.
- Semi-strong: all public information is.
- Strong: even private information is.

## What the test probes

- The variance ratio tests the weak form.
- A ratio near one supports it.
- Our 1.30 hints at a small, exploitable trend.

# The Runs Test and Its Limits

Counting streaks and what proof requires

**The runs test asks a simpler question: do up and down days alternate the way pure chance would? We count the runs - streaks of same-sign returns - and compare to the number randomness predicts, via a z-score. Then we face the deeper problem: even a clean result can't prove efficiency, because every test assumes a**

## The sequence

- Ten days of return signs.
- Five up, five down.
- Grouped into streaks.

## What we'll do by hand

- Count the runs.
- Compare to the expected number.
- Compute the z-score and judge.

# The Runs Test

Streaks against chance

$$\mathbb{E}[R] = \frac{2n_+n_-}{n} + 1 = 6, \quad z = \frac{R - \mathbb{E}[R]}{\text{sd}} = \frac{5 - 6}{1.49} = -0.67$$

## The expectation

- Under randomness, the expected runs is a formula.
- From the counts of ups and downs.
- Here it comes to six.

## The test

- Too few runs means trending.
- Too many means mean reversion.
- A z-score measures the gap.

# Counting the Runs

Observed against expected

| quantity      | value                   |
|---------------|-------------------------|
| signs         | + + + - - + - - - +     |
| runs observed | 5                       |
| expected runs | 6.0                     |
| z-score       | -0.67 (not significant) |

The ten days form five runs - three ups, two downs, one up, three downs, one up. Randomness predicts six, with a standard deviation of about 1.49, so the z-score is minus 0.67. Well inside plus-or-minus 1.96: we cannot reject weak-form efficiency.

# Why Proof Is Elusive

The joint hypothesis and the paradox

$|z| = 0.67 < 1.96 \Rightarrow$  cannot reject efficiency; but a test needs a return model (joint hypothesis)

## The joint hypothesis problem

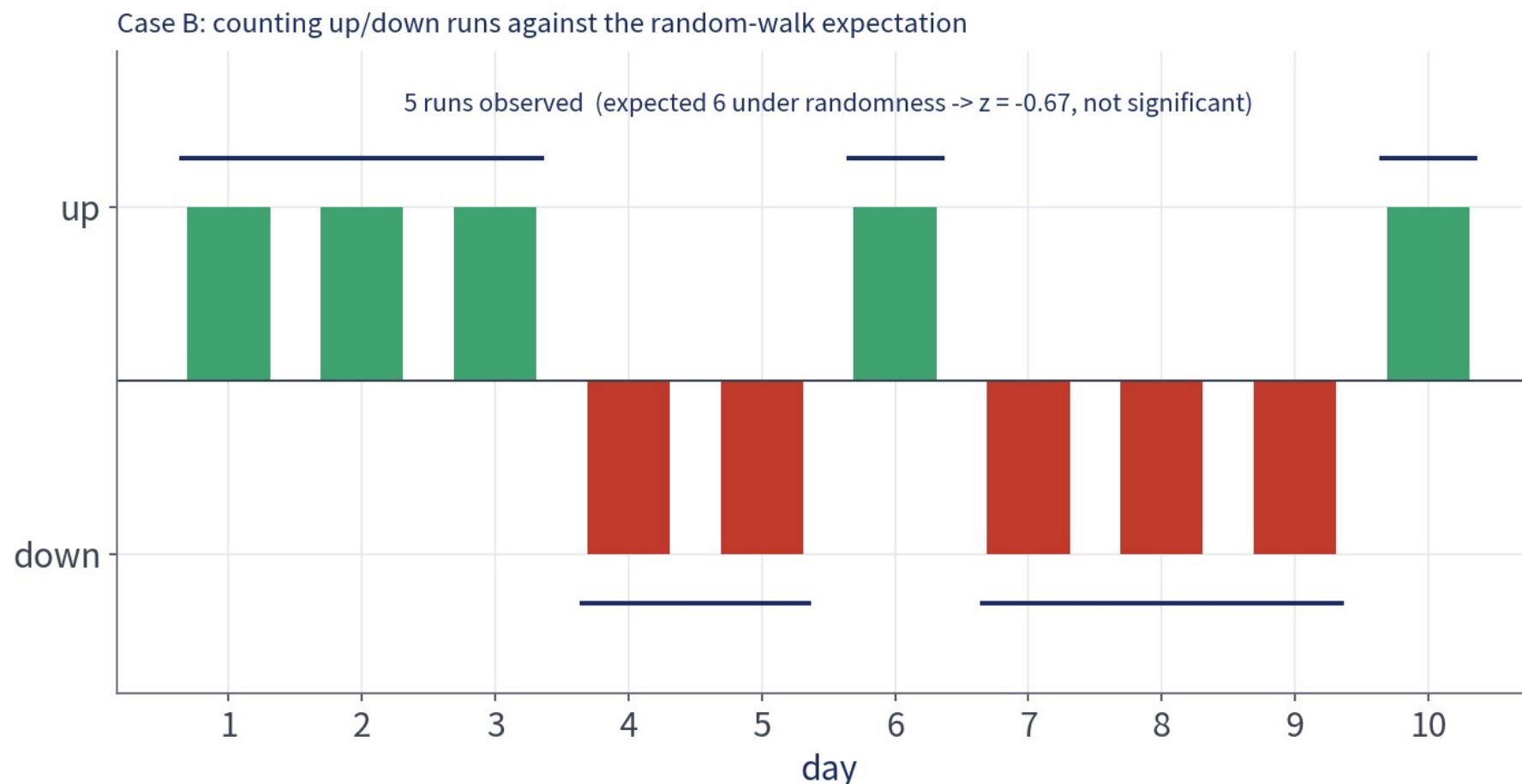
- Any test needs a model of expected returns.
- A rejection could blame either one.
- So efficiency is never tested alone.

## Grossman-Stiglitz

- If prices reflected everything, no one would research.
- But then nothing would get priced in.
- Perfect efficiency is self-defeating.

# Five Streaks, Six Expected

Up days in green, down days in red, with the runs bracketed above. Five streaks appear where pure chance would produce about six - a difference of one, small against a standard deviation of 1.49. The z-score of minus 0.67 sits comfortably inside the band: this sequence is entirely consistent with a random walk.



# Efficiency as a Moving Target

The adaptive view

## Why we rarely reject

- Markets are close to weak-form efficient.
- Direction is hard to predict - as Part 2 showed.
- Small edges are quickly arbitrated away.

## The adaptive markets view

- Efficiency evolves as participants learn.
- Opportunities appear, get competed away, recur.
- Not a fixed law, but an ecology.

# The Efficiency Toolkit

The weak-form tests share one logic: compare what you observe to what a random walk predicts. The variance ratio checks whether variance scales linearly; the runs test checks whether streaks match chance; the autocorrelation test, from Part 2, checks direct predictability. Deviations hint at trending or reversion. But interpret with humility: the joint hypothesis problem and the

## The variance ratio (Case A)

- Observed 26 over predicted 20.
- A ratio of 1.30 - trending.
- A small weak-form deviation.

## The runs test (Case B)

- Five runs against an expected six.
- z of -0.67 - not significant.
- Consistent with a random walk.

# Reproduce the Variance Ratio and Runs Test

Direct Claude Code to check your arithmetic

GOAL Compute a variance ratio, then a runs-test z-score - the two weak-form tests.

## [INPUT] Prompt · Claude Code

```
Write and run part3_practice.py that:  
CASE A - variance ratio:  
  * with a 1-day variance of 4 and an observed 5-day  
    variance of 26, print VR(5) and interpret it;  
  * confirm VR = 1 would mean a random walk.  
CASE B - runs test:  
  * for the signs + + + - - + - - +, count the runs;  
  * print the expected runs, the sd, and the z-score;  
  * report whether efficiency can be rejected.  
Then plot variance vs horizon and the sign runs.
```

## Check against the slides

- $VR(5) = 26/20 = 1.30$ .
- trending ( $VR > 1$ ).
- runs 5, expected 6.
- $z = -0.67$  (not significant).

Deliverable: part3\_practice.py whose printed numbers match the variance ratio and the runs test.

# Three Questions on the Cases

If you can answer these, the practice has landed

**Q1** What does a variance ratio above one mean?

**Positive autocorrelation - the series trends**

**Q2** What does the runs test compare?

**Observed streaks against what randomness predicts**

**Q3** Why can't efficiency be tested alone?

**Every test also assumes a model of expected returns**

All three come from the variance ratio, the runs test, and the joint hypothesis problem.

# Three Pitfalls to Avoid

Where efficiency testing trips people

## Common mistakes

- "A variance ratio above one guarantees profit."
- "Failing to reject proves the market efficient."
- "A rejection proves inefficiency."

## The reality

- Edges are small and often vanish after costs.
- Failing to reject is weak evidence, not proof.
- It might be the return model that's wrong.

# Two Calculations to Remember

Part 3 applications in one screen

1

## **The variance ratio spots trending**

With a one-day variance of 4, a random walk predicts a five-day variance of 20 - but the observed 26 gave a variance ratio of 1.30. Above one means variance grows faster than linearly: positive autocorrelation, a mild trend, a small weak-form deviation.

2

## **The runs test found nothing to reject**

The ten-day sign sequence formed five runs where randomness predicts six, a z-score of minus 0.67 - well inside the band. We cannot reject weak-form efficiency, the usual outcome: direction is genuinely hard to predict.

3

## **Efficiency is a benchmark, not a proof**

The joint hypothesis problem ties every test to a return model, and Grossman-Stiglitz shows perfect efficiency is self-defeating. Treat efficiency as a disciplined baseline - and the adaptive-markets view as how it evolves.

**Week 1 complete — returns, their stylized facts, and the efficiency that makes direction so hard to predict.**